Anand Bhattad

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Education

University of Illinois Urbana-Champaign (UIUC), USA

Ph.D., Computer Science — Jan'19 - May'24
Thesis: **Exploring Knowledge in Generative Models**
Thesis committee: **David Forsyth (Advisor)**, Alexei (Alyosha) Efros, Bill Freeman, Derek Hoiem, Svetlana Lazebnik and Shenlong Wang

M.S., Computer Science — Aug'17 - Dec'18
M.S., Civil Engineering — Aug'15 - Aug'17

National Institute of Technology Karnataka (NITK) Surathkal, India

B.Tech., Civil Engineering — July'11 - May'15

Experience

Toyota Technological Institute at Chicago, Research Assistant Professor — Sept'23 - present

University of California Berkeley, Visiting Scholar; Host: Alexei (Alyosha) Efros — May'24 - present

Allen Institute of Artificial Intelligence, Research Intern with Ani Kembhavi — Feb'23 - Apr'23

Intel, Research Intern with Stephan R. Richter in Vladlen Koltun's team — Feb'21 - Aug'21

NVIDIA, Research Intern with Andrew Tao and Bryan Catanzaro — May'20 - Dec'20

Fyusion Inc, Research Intern with Abhishek Kar and Rodrigo Ortiz Cayon — May'19 - Aug'19

Siemens Corporate Technology, Computer Vision & DL Intern with Jan Ernst — May'17 - Aug'17

Michigan State University, Research Intern with Prof. Nizar Lajnef — May'14 - Jul'14

IIIT Hyderabad, Research Intern with Prof. Ramacharla Pradeep Kumar — Jun'13 - Jul'13

Preprints/Under Review (self-selected Key Publications)

4. **Latent Intrinsics Emerge from Training to Relight**
X. Zhang, W. Gao, S. Jain, M. Maire, D.A. Forsyth, **A. Bhattad**
In Submission

3. **From an Image to a Scene: Learning to Imagine the World from a Million 360° Videos**
M. Wallingford, **A. Bhattad**, A. Kusupati, V. Ramanujan, M. Deitke, A. Kembhavi, R. Mottaghi, W. Ma, A. Farhadi
In Submission

2. **Generative Models: What do they know? Do they know things? Let's find out!**
X. Du, N. Kolkin, G. Shakhnarovich, **A. Bhattad**
In Submission

1. **Intrinsic ControlNet: Zero-shot Object Compositing from Image Intrinsics**
Z. Zhang, F. Fortier-Chouinard, M. Garon, **A. Bhattad**, JF. Lalonde
In Submission

Refereed Conference Papers (self-selected Key Publications)

11. **Videoshop: Localized Semantic Video Editing with Noise-Extrapolated Diffusion Inversion**
X. Fan, **A. Bhattad***, R. Krishna* (* for equal advising)
ECCV 2024

10. ***Shadows Don't Lie and Lines Can't Bend!* Generative Models don't know Projective Geometry...for now**
A. Sarkar*, H. Mai*, A. Mahapatra*, S. Lazebnik, D.A. Forsyth, **A. Bhattad**
CVPR 2024

9. **StyLitGAN: Image-based Relighting via Latent Control**
A. Bhattad, J. Soole, D. A. Forsyth
CVPR 2024 [rejected 3× before accepted]

8. **StyleGAN Knows Normals, Depth, Albedo, and More**
A. Bhattad, D. McKee, D. Hoiem, D. A. Forsyth
NeurIPS 2023

7. **OBJect 3DIT: Language-guided 3D-aware Image Editing**
O. Michel, **A. Bhattad**, E. Vanderbilt, R. Krishna, A. Kembhavi, T. Gupta
NeurIPS 2023

6. **Improving Equivariance in State-of-the-Art Supervised Depth and Normal Predictors**
Yuanyi Zhong, **A. Bhattad**, Y. Wang, D. A. Forsyth
ICCV 2023 [rejected 1× before accepted]

5. **Cut-and-Paste Object Insertion by Enabling Deep Image Prior for Reshading**
A. Bhattad, D. A. Forsyth
3DV 2022 ***(Spotlight Presentation)*** [rejected 3× before accepted]

4. **DIVeR: Real-time and Accurate Neural Radiance Fields with Deterministic Integration for Volume Rendering**
L. Wu , J. Y. Lee, **A. Bhattad**, Y. Wang, D. A. Forsyth
CVPR 2022 ***(Oral Presentation; Best Paper Finalist)***

3. **View Generalization for Single Image Textured 3D Models**
A. Bhattad, A. Dundar, G. Liu, A. Tao, B. Catanzaro
CVPR 2021

2. **Unrestricted Adversarial Perturbations via Semantic Manipulation**
A. Bhattad*, M. J. Chong*, K. Liang, B. Li, D. A. Forsyth (* for equal contribution)
ICLR 2020 [rejected 2× before accepted]

1. **Improved Style Transfer with Calibrated Metrics**
M. Yeh*, S. Tang*, **A. Bhattad**, C. Zou, D. A. Forsyth (* for equal contribution)
WACV 2020 [rejected 3× before accepted]

Refereed Workshop Papers

5. **Intrinsic LoRA: A Generalist Approach for Discovering Knowledge in Generative Models**
X. Du, N. Kolkin, G. Shakhnarovich, **A. Bhattad**
CVPR-W 2024 ***(Oral Presentation)***
workshops: Generative Models for CV, AI for 3D Gen, SyntaGen, SynthData4CV, Dataset Distillation

4. **UrbanIR: Large-Scale Urban Scene Inverse Rendering from a Single Video**
Z. Lin, B. Liu, Y. Chen, D.A. Forsyth, J. Huang, **A. Bhattad**, S. Wang
CVPR-W 2024
workshop: SynthData4CV

3. **MIMIC: Masked Image Modeling with Image Correspondences**
   K. Marathe, M. Bigverdi, N. A. Khan, T. Kundu, P. Howe, S. Ranjit, **A. Bhattad**, A. Kembhavi, L. Shapiro, R. Krishna
   CVPR-W 2024
   workshop: 3D with Multi-View Supervision (Archival, 8-page paper)

2. **Big but Imperceptible Adversarial Perturbations via Semantic Manipulation**
   A. Bhattad*, M. J. Chong*, K. Liang, B. Li, D. A. Forsyth (* for equal contribution)
   CVPR-W 2019 ***(Oral Presentation)*** [arXiv]
   workshop: Adversarial Machine Learning in Real-World Computer Vision Systems

1. **Detecting Anomalous Faces with "No Peeking" Autoencoders**
   A. Bhattad, J. Rock, D. A. Forsyth
   CVPR-W 2018 ***(Oral Presentation)***
   workshop: Vision with Biased and Scarce Data

Technical Reports

3. **Blocks2World: Controlling Realistic Scenes with Editable Primitives**
   V. Vavilala, S. Jain, R. Vasanth, **A. Bhattad**, D.A. Forsyth
   Technical Report, arXiv 2023

2. **Make It So: Steering StyleGAN for Any Image Inversion and Editing**
   A. Bhattad, V. Shah, D. Hoiem, D. A. Forsyth
   Technical Report, arXiv 2023

1. **SIRfyN: Single Image Relighting from your Neighbors**
   D. A. Forsyth, **A. Bhattad**, P. Asthana, Y. Zhong, Y. Wang
   Technical Report, arXiv 2021

Teaching Experience

Past Meets Present: A Tale of Two Visions, TTIC; *Course Instructor* Spring 2024

- Designed and developed a new course on Computer Vision, combining historical and modern approaches
- Taught a class of 20+ students from TTIC and UChicago, fostering an engaging learning environment

Computer Science, UIUC; *Graduate Teaching Assistant* Jan 2016 – Dec 2018

- Applied Machine Learning (CS498 AML), Fall 2018
  - Designed and implemented Kaggle competitions to enhance student learning and engagement
- Data Structures (CS 225), Spring 2017
  - Developed assignments, exams, and labs to assess and reinforce student understanding
- Introduction to Computing (CS 101), Spring 2016 & Fall 2016
  - Recognized as an **Excellent TA** for effective teaching and student support

Mentoring

Doctoral Students

- Xiao Zhang (University of Chicago; 2023-present) | (see UR#4)
- Matt Wallingford (University of Washington; 2023-present) | (see UR#3)
- Xiaodan Du (TTIC; 2023-present) | Paper at CVPR-W (see WP#1) | Paper in Review (see UR#2)
- Xiang Fan (University of Washington; 2023-present) | Paper at ECCV 2024 (see CP#11)
- Zitian Zhang (Université Laval; 2023-present) | Paper in Review (see UR#1)
- Ayush Sarkar (UIUC; 2023-present) | Paper at CVPR (see CP#10)
- Kalyani Marathe (University of Washington; 2023-present) | Paper at CVPR-W (see WP#1)
- Zhi-Hao Lin (UIUC; 2023-present) | Paper at CVPR-W (see WP#4)

Masters Students

- Hanlin (Asher) Mai (UIUC; 2023-present) | Paper at CVPR (see CP#10)
- Seemandhar Jain (UIUC; 2023-2024) | (see UR#4) | Next: PhD Student at UCSD
- Frédéric Fortier-Chouinard (Université Laval; 2023-present) | Paper in Review (see UR#1)
- James Soole (UIUC; 2023) | Paper at CVPR (see CP#9)
- Feiran Wang (UIUC; 2023) | Project on NeRF + Illumination Physics
- Kexuan (Klaus) Zou (UIUC; 2019-2020) | Next: Software Engineer at NVIDIA

Bachelors Students

- Joshua Ahn (University of Chicago; 2024-present) | Project on NeRF w/o Positional Encoding
- Amitabh Mahapatra (UIUC; 2023-present) | Paper at CVPR (see CP#10)
- Kuan-Sheng Chen (UIUC; 2023) | Project on Generative Models with Intrinsic Images
- Oscar Michel (AI2; 2023) | Paper at NeurIPS (see CP#7) | Next: PhD Student at NYU
- Liwen Wu (UIUC; 2022) | Paper at CVPR (see CP#4) | Next: PhD Student at UCSD
- Brian Chen (UIUC; 2020-2022) | REU at CMU's Robotics Institute
- Anchu Zhu (UIUC; 2018-2019) | Next: MS in CS at USC

Doctoral Committees

- Xin Yuan, PhD from University of Chicago; Advisor: Michael Maire

Invited Talks

What do Generative Image Models Know?

- IIIT Hyderabad, India; Vision Seminar — Jan 2024
- TTIC; Research@TTIC — Oct 2023
- Exactech, Inc.; Tech Talk — Oct 2023

Exploring Knowledge in Generative Models

- Stanford University in Jiajun Wu's group — June 2023
- University of Tubingen, Autonomous Vision Group — May 2023

What do Generative Models know about Geometry and Illumination?

- UC Berkeley: Vision Seminar — Apr 2023
- NVIDIA Research — Apr 2023
- MIT: Vision and Graphics Seminar — Apr 2023
- CMU: VASC Seminar — Mar 2023
- UW: Vision Seminar — Mar 2023

Learning about Light without Labeled Data

- UMD: Vision Seminar — Mar 2023
- UCSD: Pixel Cafe Seminar — Feb 2023
- TTIC: Research Talk — Feb 2023

Services & Leadership

Lead Organizer, ECCV-W on "Knowledge in Generative Models" — Sept 2024
Co-Organizer, TTIC-W on "Multimodal Artificial Intelligence " — Aug 2024
Lead Organizer, CVPR-W on "CV 20/20: A Retrospective Vision" — June 2024
Lead Organizer, CVPR-W on "Scholars & Big Models: How Can Academics Adapt?" — June 2023
Lead Organizer, UIUC Vision Workshop at Allerton — April 2023
Organizer, UIUC External Speaker Series — Jan 2021 – May 2023
Organizer, Vision Lunch (reading group) — Jan 2021 – May 2023

Area Chair, Computer Vision Conferences — July 2024 – present

- Serving as an AC for WACV 2025

Reviewer, Computer Vision Conferences Nov 2018 – present

- Serving as reviewer for CVPR, NeurIPS, ECCV and ICCV conferences and workshops.
- **Outstanding Reviewer** for ICCV 2023
- **Outstanding Emergency Reviewer** for CVPR 2021

Program Committee Jun 2019

- Adversarial Machine Learning in Real-World Computer Vision Systems, CVPR 2019 Workshop
- Security and Privacy in Machine Learning, ICML 2019 Workshop

Organizer, Vision Coffee, Urbana-Champaign May'18 - Aug'18

- Vision and Deep Learning paper/topic weekly discussion group
- Finalized agenda/topic, location and initiated discussions

Co-Founder, SchoolEngg Apr'13 - May'15

- Forum for high-school students; providing firsthand knowledge of all engineering disciplines
- Currently, renamed as *PrepLift*, an education counselling start-up in India

Co-Founder, American Society of Civil Engineers (ASCE) NITK Chapter Apr'13 - May'15

- Co-Founded the first ASCE student chapter of India
- Served as treasurer for one year and as a senior advisor for the second year.
- Coordinated with board members, faculty, and outside organizations to facilitate 10+ events